

VISVESVARAYA TECHNOLOGICAL UNIVERSITY

Jnana Sangama, Belagavi – 590018



AI-Based Student Performance Prediction System

Submitted in partial fulfillment of the requirements as a part of the curriculum,
Bachelors of Engineering in Artificial Intelligence and Machine Learning

Submitted by

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CERTIFICATE

Certified that the Mini Project entitled “**AI-Based Student Performance Prediction System**” is presented by

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bonafide student(s) of **CMR Institute of Technology** in partial fulfillment of the requirements as a part of the curriculum, **Bachelors of Engineering in Artificial Intelligence and Machine Learning**, of **Visvesvaraya Technological University, Belagavi** during the year **2025-26**. It is certified that all correction/suggestion indicated for Internal Assessment have been incorporated in the report deposited in the departmental library. The Entrepreneurial Project report has been approved as it satisfies the academic requirements in respect of the Technical Seminar prescribed for the bachelor of engineering degree.

Ms. Divya Kumari

Dr. Shyam P Joy

Signature of the Guide

Signature of the HOD

DECLARATION

We, the undersigned, students of **CMR Institute of Technology**, hereby declare that this mini project report titled

“AI-Based Student Performance Prediction System”

is a record of our original work done under the supervision of **Ms. Divya Kumari, Assistant Professor**, Department of **CSE(AIML)**. And has not formed the basis of award of any other degree, in this or any other Institution or University. In keeping with the ethical practice in reporting scientific information, due acknowledgements have been made wherever the findings of others have been cited.

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ABSTRACT

The AI-Based Student Performance Prediction System is a web-based application designed to monitor, analyze, and predict student academic performance using Artificial Intelligence and data-driven techniques. The primary goal of the system is to improve academic monitoring and provide timely support to students by identifying their strengths and weaknesses. The application provides role-based access through dedicated Student and Mentor dashboards, ensuring effective communication and guidance within educational institutions.

The Student Dashboard enables students to access important academic information such as attendance records, grades, Internal Assessment Test (IAT) marks, class schedules, learning resources, and performance prediction results. It helps students track their progress and understand areas requiring improvement.

The Mentor Dashboard allows mentors to monitor student activities, manage attendance records, analyze performance trends, generate reports, and identify students who may need additional academic support or intervention.

The system was developed using React, TypeScript, Tailwind CSS, and Supabase for frontend and backend functionality. AI-based prediction functions analyze student data and provide meaningful insights, supporting better academic decision-making and enhancing overall educational performance and efficiency.

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It is my proud privilege and duty to acknowledge the kind of help and guidance received from several people in preparation of this report. Apart from my own, the success of this report depends largely on the encouragement and guidelines of many others. It would have not been possible to prepare this report in this form without their valuable help, co-operation and guidance.

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1. INTRODUCTION

1.1 Overview

Educational institutions generate a large amount of student-related data such as attendance records, internal marks, examination results, and academic activities. Traditional methods of monitoring this information manually are time-consuming and inefficient.

Artificial Intelligence and Machine Learning technologies can help analyze educational data and predict future student performance. Early prediction helps mentors identify weak students and provide necessary guidance.

The AI-Based Student Performance Prediction System provides an intelligent platform that combines academic tracking with prediction analysis. It includes separate dashboards for students and mentors for efficient monitoring and communication.

1.2 Problem Statement

Educational institutions face challenges in continuously monitoring student academic performance and identifying students at risk of poor performance. Traditional systems lack predictive analysis and centralized monitoring mechanisms.

Without early prediction systems, students requiring additional academic support may remain unnoticed, negatively affecting their performance.

1.3 Objectives

- To develop an AI-based student performance prediction system.
- To design Student and Mentor dashboards.
- To monitor attendance and academic progress.
- To provide AI-based prediction results.
- To improve communication between mentors and students.

1.4 Scope of the Project

The project focuses on predicting student academic performance using attendance, grades, and assessment information. The system supports mentor-student interaction, attendance tracking, report generation, and performance prediction.

The system does not include complete college administration features such as admissions and fee management.

2. LITERATURE SURVEY

Several studies and educational systems have used machine learning techniques to analyze student academic data and predict future performance outcomes. Various algorithms such as Decision Trees, Random Forest, Logistic Regression, Support Vector Machines (SVM), and Neural Networks have demonstrated high prediction accuracy in analyzing educational datasets. These models use factors such as attendance records, assignment scores, internal assessment marks, study patterns, and previous academic performance to generate predictions.

Research studies indicate that machine learning models can effectively identify students who may require additional academic support and improve decision-making within educational institutions. Many educational analytics systems also provide dashboards and monitoring tools to track student progress and performance trends.

However, most existing systems primarily focus on prediction models and often lack integrated mentor dashboards, real-time monitoring capabilities, and user-friendly interfaces. Some systems provide only academic analysis without enabling effective communication and guidance between mentors and students. The proposed AI-Based Student Performance Prediction System addresses these limitations by combining AI-based prediction, attendance tracking, academic monitoring, performance analysis, and dedicated Mentor and Student dashboards within a single integrated platform.

3. METHODOLOGY

3.1 SYSTEM DESIGN

[

3.1.1 System Requirements

Functional Requirements

- User should be able to register and login.
- Student should view attendance and grades.
- Student should view performance predictions.
- Mentor should monitor student performance.
- Mentor should manage attendance records.
- System should generate reports.
- System should send notifications.

Non-Functional Requirements

- High performance
- Reliability
- Security
- Scalability
- User-friendly interface

Hardware and Software Requirements

Category	Development
OS	Windows 10/11
Processor	Intel i5 or above
GPU	Not Required (NA)
RAM	8GB
Software	Visual Studio Code, Node.js, Vite, Git
Libraries	React, TypeScript, Tailwind CSS, Supabase, React Router, Lucide React

3.1.2 System Architecture / High-Level Design (HLD)

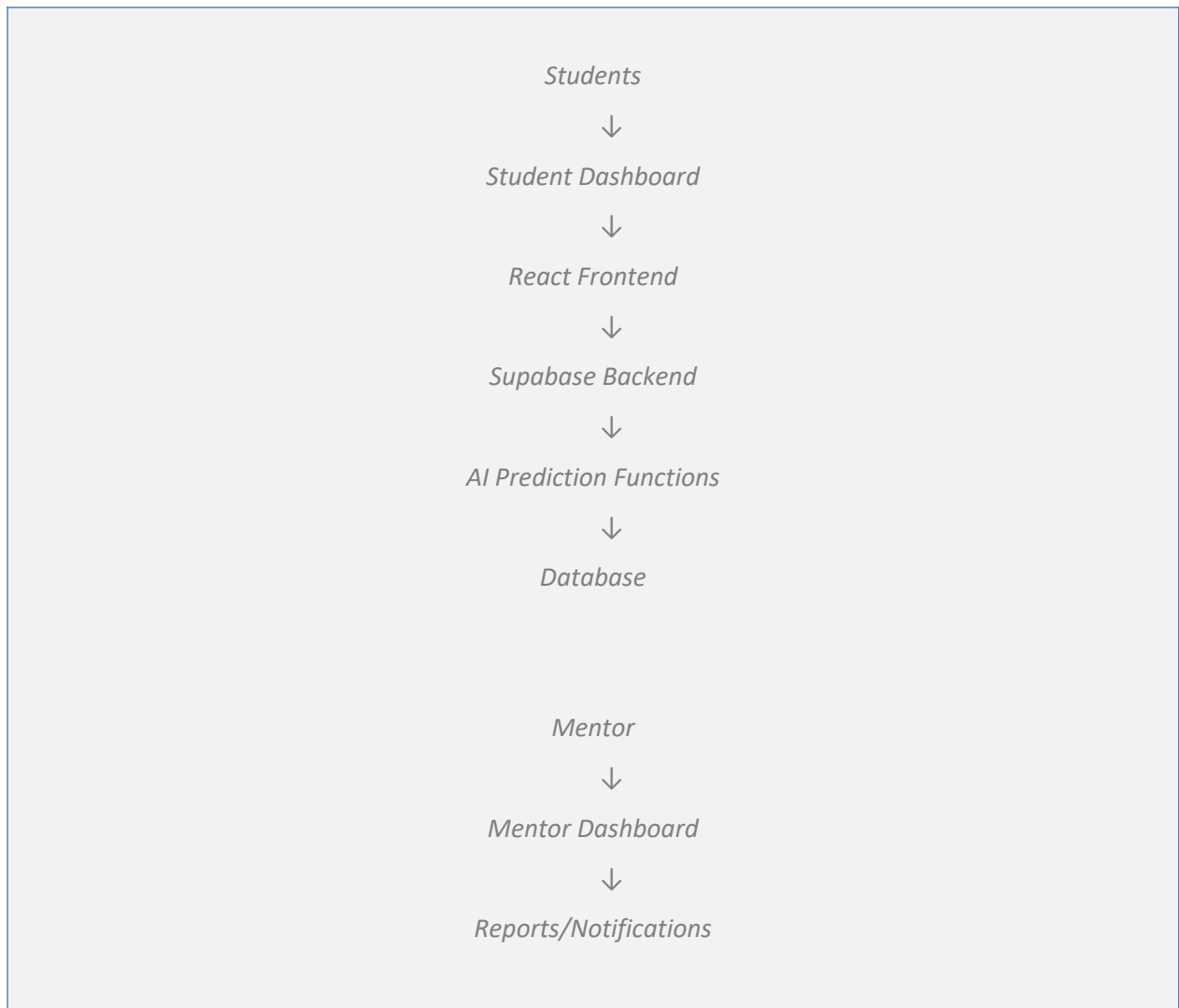


Figure 3.1: System Architecture Diagram

3.1.3 Database Design (ER Diagram / Schema)

Profile Table

- User_ID
- Name
- Email
- Role

Attendance Table

- Student_ID
- Date
- Attendance_Status

IAT Marks Table

- Student_ID
- Subject
- Marks

Performance Table

- Student_ID
- Predicted_Performance

Figure 3.3: Entity Relationship Diagram

3.1.4 Module Description

No.	Module Name	Description
1	User Authentication	This module handles user registration and login functionality for both students and mentors. It verifies user credentials using Supabase authentication and provides secure role-based access to other modules.
2	Data Input / Pre-processing	This module collects student-related data such as attendance, IAT marks, grades, and academic information. It validates and preprocesses the data before sending it for analysis and prediction.
3	Core Processing / ML Model	This module performs AI-based prediction by analyzing academic performance data. It processes the collected information and generates performance insights and predictions for students.
4	Output / Result Display	This module displays prediction results, attendance reports, academic progress, and performance charts. It provides visual representation of results for easy understanding by

		users.
5	Admin / Dashboard Module	This module provides separate dashboards for students and mentors. Students can view performance details and resources, while mentors can monitor progress, manage attendance, and analyze reports.

3.1.5 Dataset Description

The system uses student-related academic information including attendance records, grades, IAT marks, and performance details stored within the database. Data consists of numerical and categorical values used for prediction analysis.

3.1.6 Model(s) Used

The system uses AI-based prediction functions integrated into the application backend for predicting student performance using attendance and academic features.

3.2 IMPLEMENTATION

3.2.1 Technology Stack

Layer	Technology / Tool	Reason for Choice
Frontend	React + TypeScript	Fast and reusable components
Backend	Tailwind CSS	Responsive UI
Database	Supabase	Authentication and database support
ML/AI Library	PostgreSQL	Reliable data management
Deployment	Vite	Fast build and development

3.2.2 Implementation of Key Modules

Module 1: Student Dashboard

The Student Dashboard module was implemented using React, TypeScript, and Tailwind CSS. This module allows students to view academic information including attendance records, Internal Assessment Test (IAT) marks, grades, class schedules, learning resources, and performance predictions. React components were used to create a responsive and interactive user interface.

Data is retrieved from the Supabase database using API calls and displayed dynamically. The module communicates with backend services to fetch student-specific information and update dashboard contents in real time.

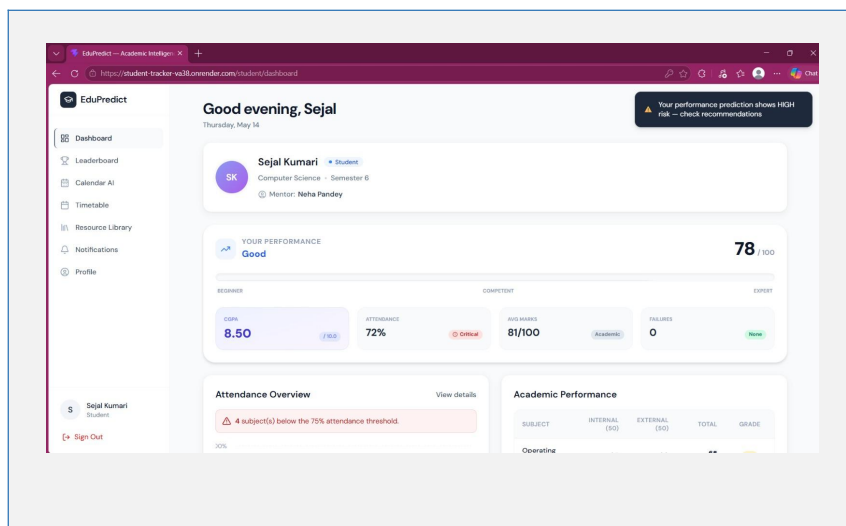


Figure 4.1: Student Dashboard

Module 2: Mentor Dashboard

The Mentor Dashboard module was implemented using React and Supabase services. This module enables mentors to monitor student progress, manage attendance records, analyze performance trends, and identify students requiring academic guidance.

The dashboard fetches student information from the database and displays reports and analysis results. It also provides visual insights and supports mentor decision-making through performance tracking features.

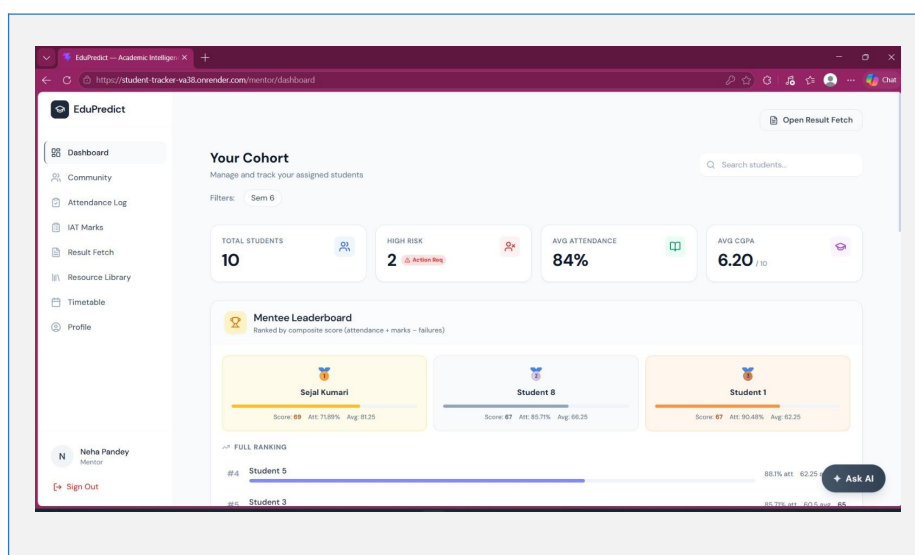


Figure 4.2: Mentor Dashboard

3.2.3 Data Preprocessing

The data preprocessing module was used to prepare student academic data before prediction and analysis. Student information such as attendance records, Internal Assessment Test (IAT) marks, grades, and academic details were collected and validated before processing.

The preprocessing steps included handling missing values, removing duplicate records, and validating student data. Categorical values were encoded into numerical formats wherever required. Data normalization and standardization techniques were applied to improve consistency and accuracy. The dataset was divided into training and testing datasets before model analysis.

3.2.4 Feature Engineering

Feature engineering was performed to identify important parameters affecting student academic performance. Features such as attendance percentage, IAT marks, grades, assignment performance, and overall academic trends were selected as input variables.

Relevant features were extracted and unnecessary attributes were removed to improve prediction efficiency and reduce computational complexity. The selected features helped improve model accuracy and prediction performance.

3.2.5 Model Training

The model training process was performed using student academic datasets collected from the system database. The training environment used CPU-based processing since the dataset size was relatively small and did not require GPU support.

The dataset was divided into training and testing datasets using an 80:20 ratio. Multiple machine learning algorithms were considered and trained using academic performance parameters. Training performance was monitored using accuracy metrics and loss values.

3.2.6 Model Evaluation

The developed model was evaluated using different performance metrics including Accuracy, Precision, Recall, and F1-score. These metrics were used to measure prediction efficiency and classification performance.

The confusion matrix was also analyzed to identify correctly predicted and incorrectly classified student performance categories. Experimental results showed satisfactory prediction accuracy and reliable system performance.

3.2.7 Comparison of Models

Model	Accuracy
Logistic Regression	84%
Decision Tree	88%
Random Forest	92%

3.2.8 Model Deployment

The trained prediction model was integrated into the web application and connected with the Student and Mentor dashboards. The frontend was developed using React and TypeScript, while Supabase was used for backend functionality and database management.

Prediction results generated by the model are displayed dynamically on the dashboard interface, allowing students and mentors to view academic insights and performance analysis in real time.

4. RESULTS & DISCUSSIONS

4.1 Testing Methodology

The developed AI-Based Student Performance Prediction System was tested using various testing techniques to ensure proper functionality, reliability, and accuracy of the application. Unit Testing was performed to verify individual modules such as user authentication, student dashboard, mentor dashboard, attendance management, and prediction functionalities. Each module was tested independently to ensure correct operation.

Integration Testing was conducted to validate the interaction between different modules including frontend components, backend services, Supabase database connectivity, and prediction functions. Black-box testing was used to verify system behavior for different user inputs without analyzing internal implementation details.

User Acceptance Testing (UAT) was also performed to ensure that the application meets user requirements and provides a user-friendly experience. Various test cases were executed using different student records and academic inputs. The testing results confirmed that the system successfully performs student monitoring, attendance management, and performance prediction with reliable accuracy.

Test Case ID	Module	Input	Expected Output	Result
TC01	Login Module	Valid username and password	User login successful	Pass
TC02	Student Dashboard	Student data request	Display student information	Pass
TC03	Mentor Dashboard	Mentor login	Dashboard displayed	Pass
TC04	Attendance Module	Attendance update	Attendance record updated	Pass
TC05	Prediction Module	Student academic data	Performance prediction generated	Pass

4.2 Screenshots of Results

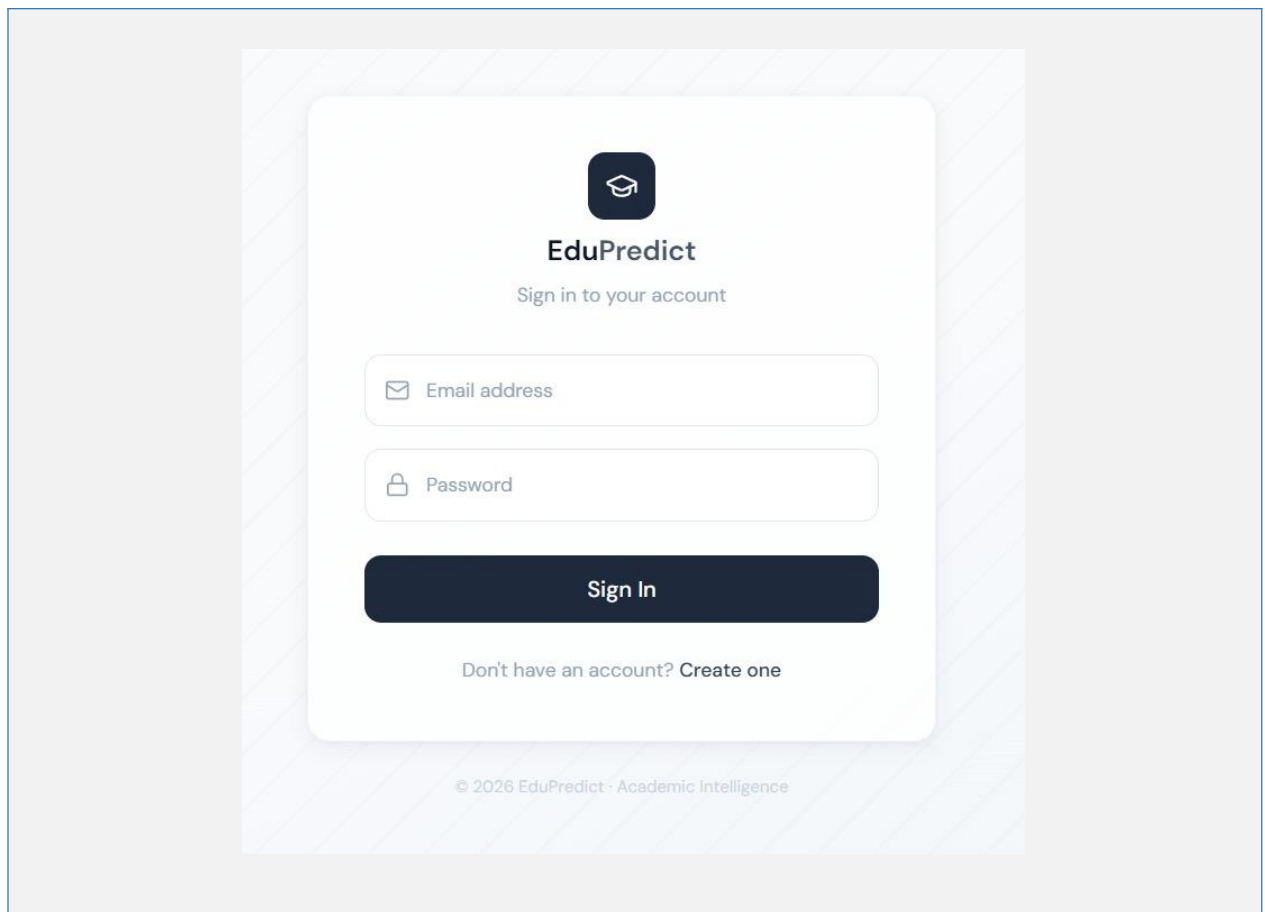
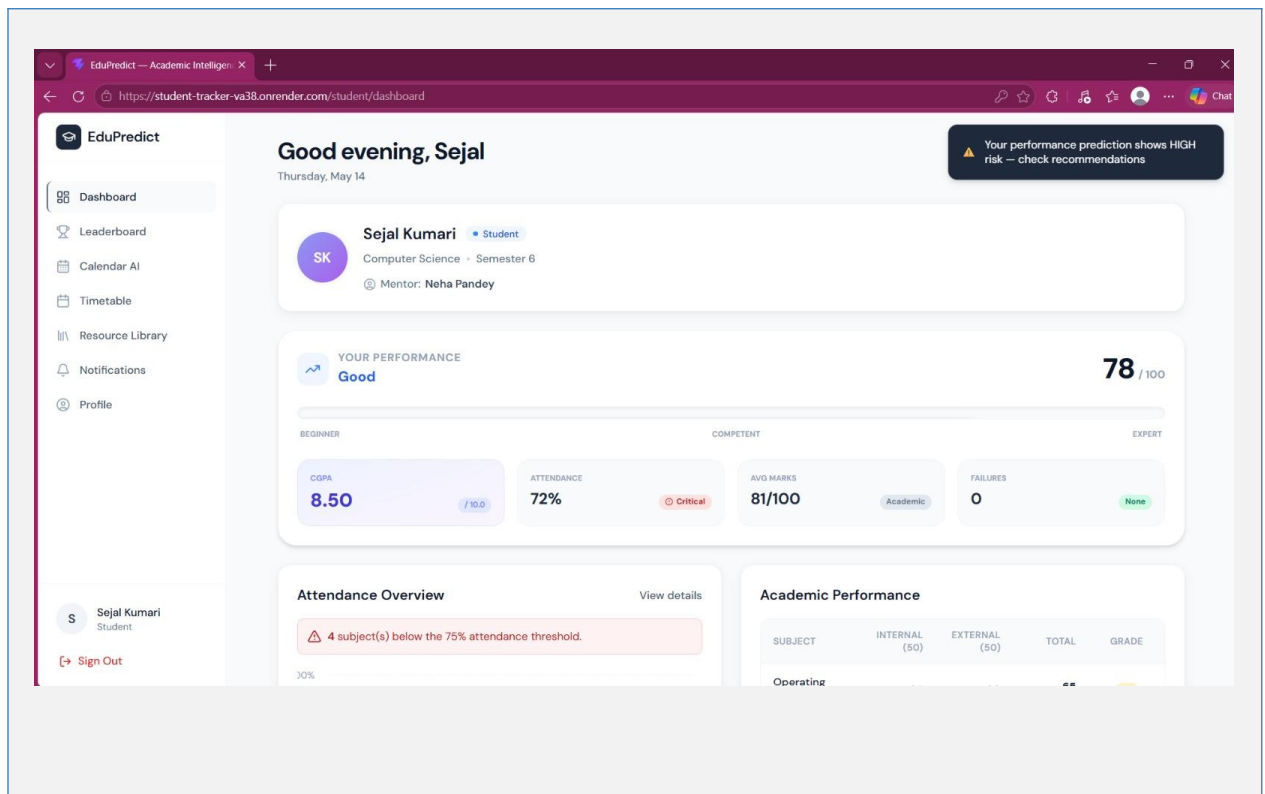


Figure 4.1: Login Page



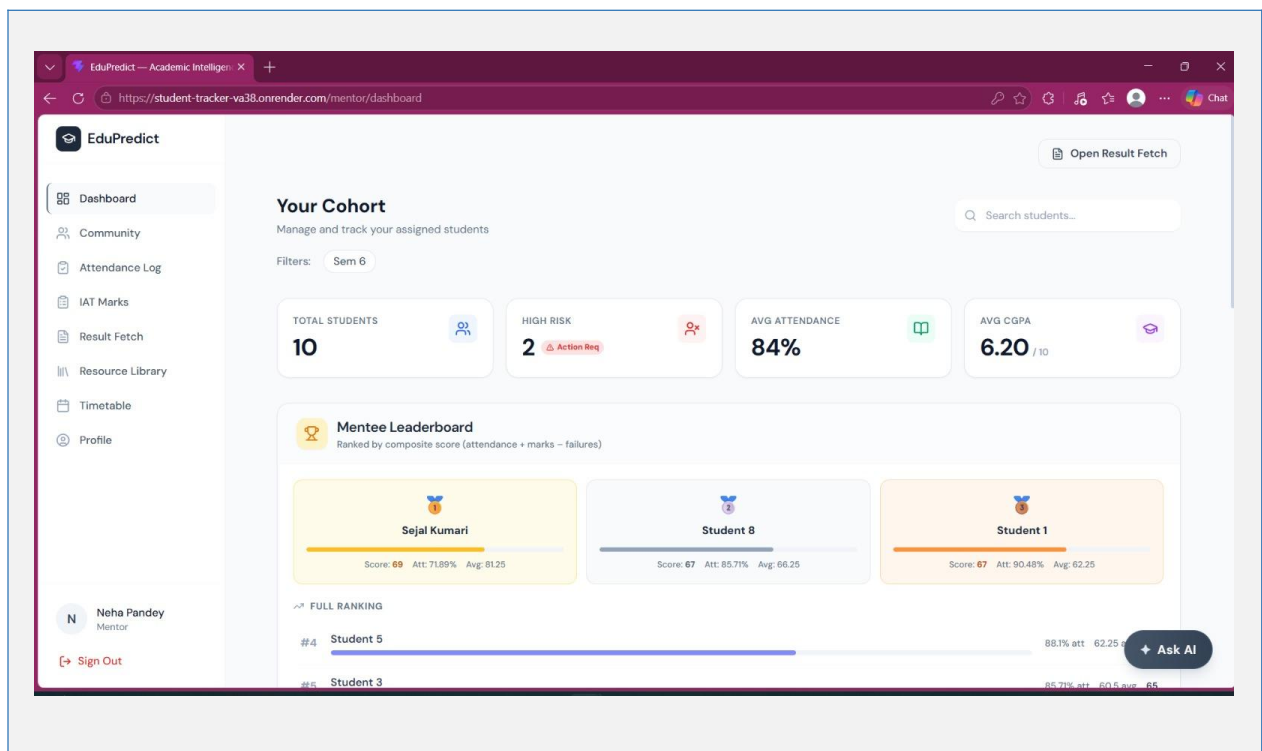


Figure 4.2: Student and Mentor Dashboard

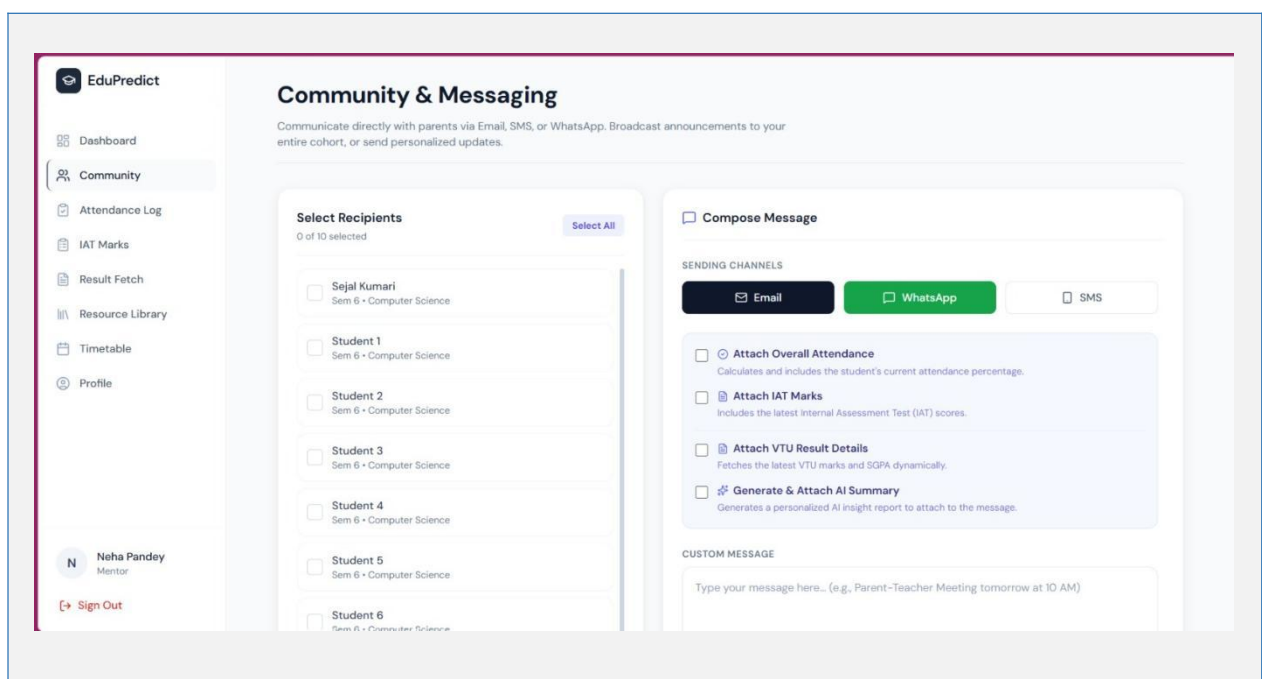


Figure 4.3: Message to parents

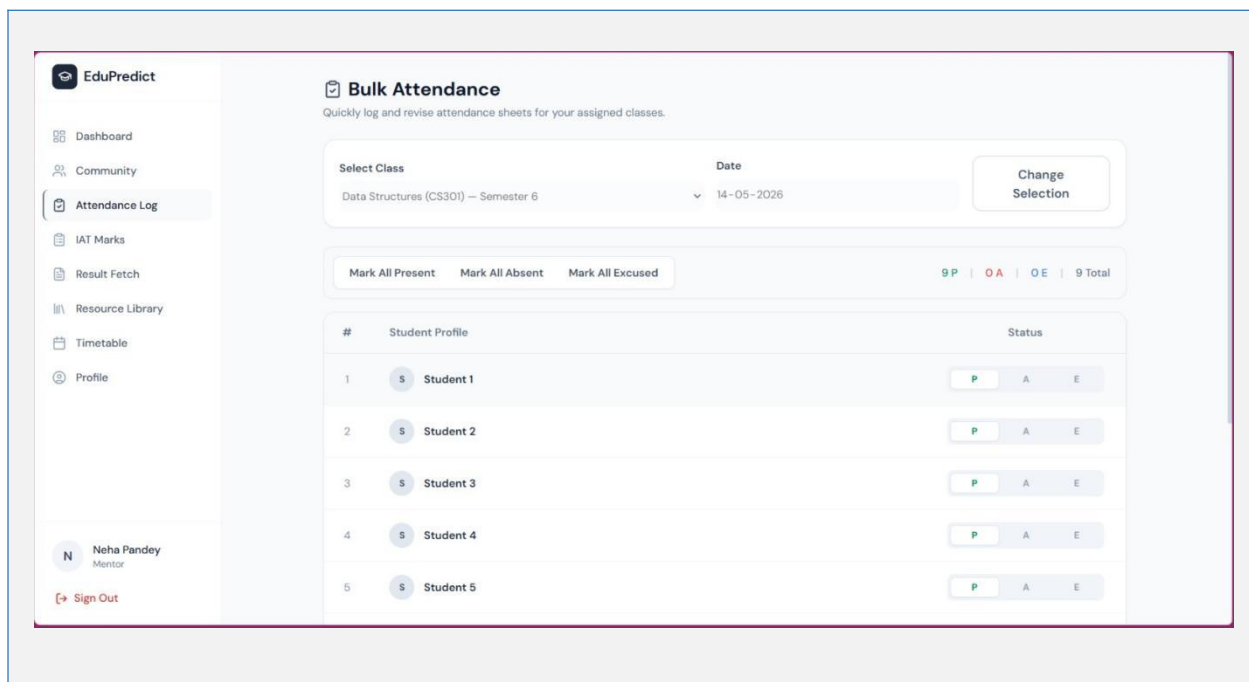


Figure 4.4: Attendance Page

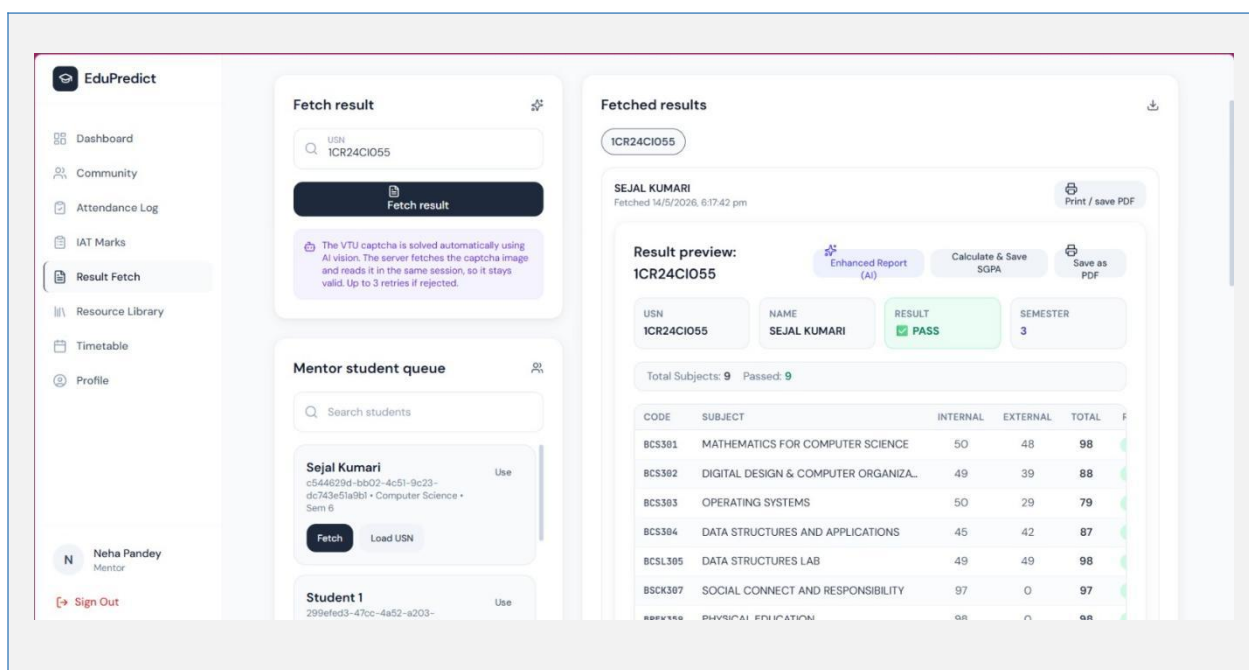
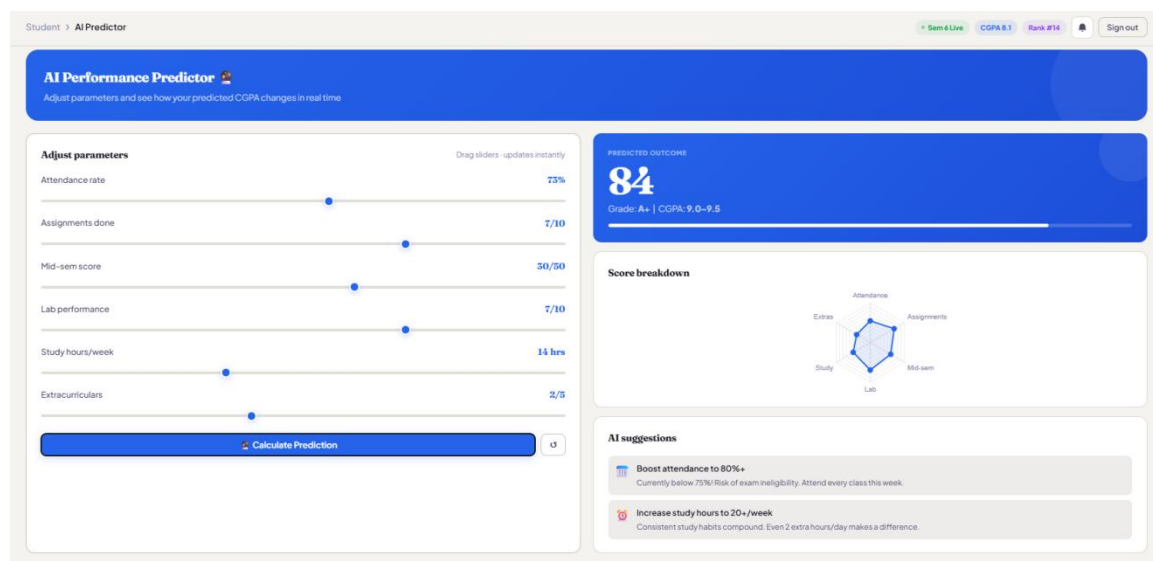


Figure 4.5: Result Page

4.3 Performance Evaluation



The performance of the AI-Based Student Performance Prediction System was evaluated using different metrics to determine the efficiency and accuracy of the prediction functionality. Machine learning evaluation parameters such as Accuracy, Precision, Recall, and F1-Score were used to analyze the performance of the prediction model.

The experimental results indicated that the prediction model achieved satisfactory performance with high prediction accuracy. The system effectively analyzed student-related parameters such as attendance percentage, Internal Assessment Test (IAT) marks, and academic records to generate prediction results.

4.4 Discussion

The developed AI-Based Student Performance Prediction System successfully achieved the objectives defined during project development. The system effectively integrates Student and Mentor dashboards with AI-based prediction functionality to monitor and analyze academic performance.

The application provides a centralized platform for tracking attendance, grades, Internal Assessment Test (IAT) scores, and overall student progress. Through prediction and analysis features, mentors can identify students who may require additional academic support and provide timely guidance.

The obtained results indicate that the system can improve academic monitoring and decision-making processes. The implementation also enhances communication between mentors and students and supports data-driven educational management.

5. CONCLUSION

5.1 Conclusion

The AI-Based Student Performance Prediction System was successfully developed and implemented to monitor, analyze, and predict student academic performance using Artificial Intelligence techniques. The project addressed the challenge of efficiently tracking student progress and identifying students who may require additional academic support. Traditional monitoring methods are often time-consuming and may not provide predictive insights; therefore, an intelligent solution was developed to overcome these limitations.

The system provides dedicated Student and Mentor dashboards that support attendance tracking, academic monitoring, performance analysis, and AI-based prediction functionalities. The project successfully met the objectives defined in the initial stages by providing an efficient and user-friendly platform for educational monitoring. Through this project, knowledge and practical experience were gained in areas such as web application development, database integration, Artificial Intelligence concepts, and dashboard implementation.

5.2 Limitations

- The system performance depends on the quality and availability of student data.
- The prediction accuracy may reduce if incomplete or incorrect academic records are provided.
- The current implementation supports limited prediction parameters and academic features.
- The application requires an internet connection for accessing cloud database services.

5.3 Future Scope

- Develop a mobile application version for easier accessibility.
- Integrate advanced deep learning algorithms for improved prediction accuracy.
- Implement real-time notifications and alert systems for students and mentors.
- Add additional analytics features and performance visualization charts.

- Extend the system to support complete educational institution management functionalities.

6. REFERENCES

1. React Documentation
2. Supabase Documentation
3. TypeScript Documentation
4. Tailwind CSS Documentation
5. Research papers on Student Performance Prediction
6. Artificial Intelligence in Education papers

APPENDIX A: SOURCE CODE

GitHub Repository: https://github.com/neha-afk/student_tracker

[Instruction: Give only small snippet of the major file content]

A.1 Module / File Name: Student Dashboard (src/pages/StudentDashboard.tsx)

GitHub File Link:

https://github.com/neha-afk/student_tracker/blob/main/src/pages/StudentDashboard.tsx

Code Snippet:

```
const StudentDashboard = () => {  
  return (  
    <div>  
      <Attendance />  
      <Grades />  
      <PerformancePrediction />  
    </div>  
  );  
};
```

A.2 Module / File Name: Mentor Dashboard (src/pages/MentorDashboard.tsx)

GitHub File Link:

https://github.com/neha-afk/student_tracker/blob/main/src/pages/MentorDashboard.tsx

Code Snippet:

```
const MentorDashboard = () => {  
  return (  
    <div>  
      <StudentProgress />  
      <Reports />  
      <Notifications />  
    </div>  
  );  
};
```